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Start Smart: Be a Better Problem Solver

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Study Guide and Review

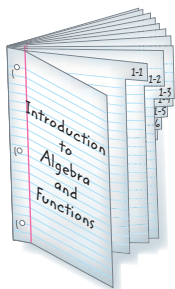


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Key Concepts

Squares and Square Roots (Lesson 1-3)

- The square of a number is the product of a number and itself.
- A square root of a number is one of its two equal factors.

Order of Operations (Lesson 1-4)

- Do all operations within grouping symbols first. Evaluate all powers before other operations. Multiply and divide in order from left to right. Add and subtract in order from left to right.

Properties (Lesson 1-8)

- Distributive Property
 $5(2 + 4) = 5 \cdot 2 + 5 \cdot 4$
 $(3 + 2)4 = 3 \cdot 4 + 2 \cdot 4$
- Commutative Property
 $3 + 2 = 2 + 3$
 $7 \cdot 4 = 4 \cdot 7$
- Associative Property
 $6 + (3 + 8) = (6 + 3) + 8$
 $5 \cdot (2 \cdot 3) = (5 \cdot 2) \cdot 3$
- Identity Property
 $4 + 0 = 4$
 $4 \cdot 1 = 4$

Functions (Lesson 1-10)

- A function is a relationship that assigns exactly one *output* value for each *input* value.
- In a function, the function rule gives the operation to perform on the input.

Key Vocabulary

algebra (p. 44)	function rule (p. 63)
algebraic expression (p. 44)	numerical expression (p. 38)
arithmetic sequence (p. 57)	order of operations (p. 38)
base (p. 30)	perfect square (p. 34)
coefficient (p. 45)	powers (p. 30)
defining the variable (p. 50)	radical sign (p. 35)
domain (p. 63)	range (p. 63)
equation (p. 49)	sequence (p. 57)
equivalent expressions (p. 53)	solution (p. 49)
evaluate (p. 31)	square (p. 34)
exponent (p. 30)	square root (p. 35)
factors (p. 30)	term (p. 57)
function (p. 63)	variable (p. 44)

Vocabulary Check

State whether each sentence is *true* or *false*. If *false*, replace the underlined word or number to make a true sentence.

1. Numerical expressions have the same value.
2. Two or more numbers that are multiplied together are called powers.
3. The range of a function is the set of input values.
4. A function assigns exactly two output values for each *input* value.
5. An equation is a sentence that contains an equals sign.
6. A sequence is an ordered list of numbers.
7. The product of a number and itself is the square root of the number.

Study Guide and Review

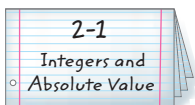


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Key Concepts

Absolute Value (Lesson 2-1)

- The absolute value of a number is the distance the number is from zero on a number line.

Comparing and Ordering Integers (Lesson 2-2)

- When two numbers are graphed on a number line, the number to the left is always less than the number to the right.

Graphing Points (Lesson 2-3)

- On a coordinate plane, the horizontal number line is the x -axis and the vertical number line is the y -axis. The origin is at $(0, 0)$ and is the point where the number lines intersect. The x -axis and y -axis separate the plane into four quadrants.

Integer Operations (Lessons 2-4, 2-5, 2-6, 2-8)

- To add integers with the same sign, add their absolute value. The sum is positive if both integers are positive and negative if both integers are negative.
- The sum of any number and its additive inverse is 0.
- To add integers with different signs, subtract their absolute values. The sum is positive if the positive integer's absolute value is greater and negative if the negative integer's absolute value is greater.
- To subtract an integer, add its opposite.
- The product or quotient of two integers with different signs is negative.
- The product or quotient of two integers with the same sign is positive.

Key Vocabulary

absolute value (p. 81)	origin (p. 88)
additive inverse (p. 96)	positive integer (p. 80)
coordinate plane (p. 88)	quadrant (p. 88)
graph (p. 80)	x -axis (p. 88)
integer (p. 80)	x -coordinate (p. 88)
negative integer (p. 80)	y -axis (p. 88)
opposites (p. 96)	y -coordinate (p. 88)
ordered pair (p. 88)	

Vocabulary Check

State whether each sentence is *true* or *false*.
If *false*, replace the underlined word or number to make a true sentence.

1. Integers less than zero are positive integers.
2. The origin is the point where the x -axis and y -axis intersect.
3. The absolute value of 7 is -7 .
4. The sum of two negative integers is positive.
5. The x -coordinate of the ordered pair $(2, -3)$ is -3 .
6. Two integers that are opposites are also called additive inverses.
7. The product of a positive and a negative integer is negative.
8. The x -axis and the y -axis separate the plane into four coordinates.
9. The quotient of two negative integers is negative.

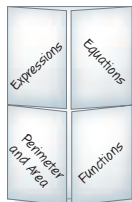
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Key Concepts

Solving Equations (Lessons 3-2, 3-3, and 3-5)

- If you add or subtract the same number from each side of an equation, the two sides remain equal.
- If you divide each side of an equation by the same nonzero number, the two sides remain equal.
- To solve a two-step equation like $3x + 4 = 19$ or $2x - 1 = -5$:

Step 1 Undo the addition or subtraction first.

Step 2 Then undo the multiplication or division.

Perimeter and Area Formulas (Lesson 3-6)

- The perimeter P of a rectangle is twice the sum of the length ℓ and width w .
- The area A of a rectangle is the product of the length ℓ and width w .

Linear Functions (Lesson 3-7)

- The graph of a linear function is a straight line. Ordered pairs in the form (x, y) can be used to represent a function and graphed on the coordinate plane as part of the graph of the function.

Key Vocabulary

formula (p. 144)

linear equation (p. 164)

two-step equation (p. 151)

work backward strategy (p. 148)

Vocabulary Check

State whether each sentence is *true* or *false*.
If *false*, replace the underlined word or
number to make a true sentence.

1. The expression $\frac{1}{3}y$ means one-third of y .
2. The words *more than* sometimes suggest the operation of multiplication.
3. The formula $d = rt$ gives the distance d traveled at a rate of r for t units of time.
4. The algebraic expression representing the words *six less than m* is $6 - m$.
5. Use the work backward strategy when you are given a final result and asked to find an earlier amount.
6. The word *each* sometimes suggests the operation of division.
7. In solving the equation $4x + 3 = 15$, first divide each side by 4.
8. The solution to the equation $p + 4.4 = 11.6$ is 7.2.
9. The process of solving a two-step equation uses the work backward strategy.
10. The expression $5x$ means 5 more than x .
11. To find the distance around a rectangle, use the formula for its area.
12. The word *per* sometimes suggests the operation of subtraction.

Study Guide and Review

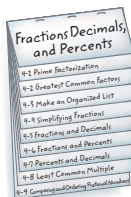


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Key Concept

Greatest Common Factor (Lesson 4-2)

- The greatest common factor or GCF is the greatest of the common factors of two or more numbers.

Fractions, Decimals, and Percents

(Lessons 4-4 to 4-7)

- A fraction is in simplest form when the GCF of the numerator and denominator is 1.
- A terminating decimal is a decimal whose digits end. Repeating decimals have a pattern in the digits that repeats forever.
- A percent is a part to whole ratio that compares a number to 100.
- To write a percent as a decimal, divide the percent by 100 and remove the percent symbol.
- To write a decimal as a percent, multiply the percent by 100 and add the percent symbol.

Least Common Multiple (Lesson 4-8)

- The least common multiple or LCM of two or more numbers is the least of their common multiples.

Rational Numbers (Lesson 4-9)

- A rational number is one that can be expressed as a fraction.

Key Vocabulary

bar notation (p. 197)	multiple (p. 211)
common denominator (p. 216)	percent (p. 202)
composite number (p. 181)	prime factorization (p. 182)
equivalent fractions (p. 192)	prime number (p. 181)
factor tree (p. 182)	ratio (p. 202)
greatest common factor (GCF) (p. 186)	rational number (p. 215)
least common denominator (p. 216)	repeating decimal (p. 197)
least common multiple (LCM) (p. 211)	simplest form (p. 192)
	terminating decimal (p. 197)

Vocabulary Check

State whether each sentence is *true* or *false*.
If *false*, replace the underlined word or number to make a true sentence.

1. A ratio is a comparison of two numbers by multiplication.
2. A rational number is a whole number greater than 1 that has exactly two factors, 1 and itself.
3. 1.875 is an example of a terminating decimal.
4. A common denominator for the fractions $\frac{2}{3}$ and $\frac{1}{4}$ is 12.
5. The greatest common factor of 3 and 5 is 15.
6. A ratio that compares a number to 100 is a percent.
7. The fractions $\frac{9}{21}$ and $\frac{3}{7}$ are equivalent fractions.

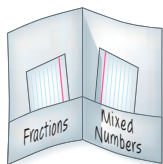
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Key Concepts

Estimating with Fractions (Lesson 5-1)

- When the numerator is much smaller than the denominator, round the fraction to 0.
- When the numerator is about half of the denominator, round the fraction to $\frac{1}{2}$.
- When the numerator is almost as large as the denominator, round the fraction to 1.

Adding and Subtracting Fractions

(Lessons 5-2 and 5-3)

- To add or subtract like fractions, add or subtract the numerators and write the result over the denominator.
- To add or subtract unlike fractions, rename the fractions using the LCD. Then add or subtract as with like fractions.
- To add or subtract mixed numbers, first add or subtract the fractions. If necessary, rename them using the LCD. Then add or subtract the whole numbers and simplify if necessary.

Multiplying and Dividing Fractions

(Lessons 5-5 and 5-7)

- To multiply fractions, multiply the numerators and multiply the denominators.
- The product of a number and its multiplicative inverse is 1.
- To divide by a fraction, multiply by its multiplicative inverse, or reciprocal.

Solving Equations (Lesson 5-6)

If you multiply each side of an equation by the same nonzero number, the two sides remain equal.

Key Vocabulary

compatible numbers (p. 232)	multiplicative inverse (p. 258)
like fractions (p. 236)	reciprocal (p. 258)
	unlike fractions (p. 237)

Vocabulary Check

Choose the correct term or number to
complete each sentence.

- To add like fractions, add the (numerators, denominators).
- The symbol $\approx$ means is (*approximately*, *exactly*) equal to.
- When dividing by a fraction, multiply by its (value, reciprocal).
- When estimating, if the numerator of a fraction is much smaller than the denominator, round the fraction to $(0, \frac{1}{2})$.
- Fractions with different denominators are called (like, unlike) fractions.
- The multiplicative inverse of $\frac{5}{6}$ is $(\frac{6}{5}, -\frac{5}{6})$.
- The mixed number $2\frac{4}{7}$ can be renamed as $(2\frac{7}{7}, 1\frac{11}{7})$.
- When multiplying fractions, multiply the numerators and (multiply, keep) the denominators.
- The reciprocal of $\frac{1}{3}$ is $(-3, 3)$.
- The fractions $\frac{4}{16}$ and $\frac{2}{4}$ are (like, unlike) fractions.
- Another word for multiplicative inverse is (reciprocal, denominator).
- The fraction $\frac{x}{2}$ can be read x multiplied by $(2, \frac{1}{2})$.

Study Guide and Review

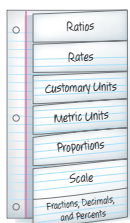


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Key Concepts

Ratios and Rates (Lessons 6-1 and 6-2)

- A ratio is a comparison of two quantities by division.
- A rate is a ratio comparing two quantities with different kinds of units.

Changing Customary Units (Lesson 6-3)

- To convert from larger units to smaller units, multiply by the appropriate unit ratio.
- To convert from smaller units to larger units, multiply by the reciprocal of the appropriate unit ratio.

Changing Metric Units (Lesson 6-4)

- When converting metric units, multiply by the appropriate power of 10.
- When converting between customary and metric units, use the appropriate unit ratio or relationship.

Proportions (Lesson 6-5)

- Proportions are equations that state two ratios or rates are equivalent.

Scale Drawings (Lesson 6-7)

- Scale drawings represent something that is too large or too small to be drawn at actual size.

Key Vocabulary

cross products (p. 306)	scale (p. 316)
equivalent ratios (p. 283)	scale drawing (p. 316)
proportion (p. 300)	scale factor (p. 318)
proportional (p. 306)	scale model (p. 316)
rate (p. 287)	unit rate (p. 287)
ratio (p. 282)	unit ratio (p. 295)

Vocabulary Check

Choose the term from the list above that best matches each phrase.

1. a comparison of two quantities by division
2. two ratios that have the same value
3. a ratio of two measurements with different units
4. an equation that shows that two ratios or rates are equivalent
5. used to represent something that is too large or too small for an actual-size drawing
6. the ratio of the distance on a map to the actual distance
7. a scale written as a ratio in simplest form
8. product of the numerator of one ratio in a proportion and the denominator of the other ratio
9. a rate that is simplified so that it has a denominator of 1
10. two quantities that have a constant rate or ratio

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7-1	7-2
7-3	7-4
7-5	7-6
7-7	7-8

Key Concepts

Percent of a Number (Lesson 7-1)

- To find the percent of a number, first write the percent as either a fraction or decimal and then multiply.

Percent Proportion (Lesson 7-2)

$$\frac{\text{part}}{\text{whole}} = \frac{n}{100} \text{ } \left. \vphantom{\frac{\text{part}}{\text{whole}}} \right\} \text{percent}$$

Percent and Estimation (Lesson 7-3)

- One way to estimate the percent of a number is to use a fraction. The other way is to first find 10% of the number and then multiply.

Percent Equation (Lesson 7-5)

$$\text{part} = \text{percent} \cdot \text{whole}$$

Percent of Change (Lesson 7-6)

- A percent of change is ratio that compares the change in quantity to the original amount.

$$\text{percent of change} = \frac{\text{amount of change}}{\text{original amount}}$$

Sales and Discount (Lesson 7-7)

- Sales tax is an additional amount of money charged on items. The total cost of an item is the regular price plus the sales tax.
- Discount is the amount by which the regular price of an item is reduced. The sale price is the regular price minus the discount.

Simple Interest (Lesson 7-8)

- Simple interest is the amount paid or earned for the use of money.

$$I = prt$$

Key Vocabulary

discount (p. 375) percent proportion (p. 350)
percent equation (p. 361) principal (p. 379)
percent of change (p. 369) sales tax (p. 375)
percent of decrease (p. 369) simple interest (p. 379)
percent of increase (p. 369)

Vocabulary Check

State whether each sentence is *true* or *false*.
If *false*, replace the underlined word or
number to make a true sentence.

- The sale price of a discounted item is the regular price minus the discount.
- A ratio that compares the change in quantity to the original amount is called the percent of change.
- A percent proportion compares part of a quantity to the whole quantity using a percent.
- The formula for simple interest is $I = prt$.
- A method for estimating the percent of a number is to find 21% of the number and then multiply.
- The equation $\text{part} = \text{percent} \cdot \text{whole}$ is known as the principal equation.
- The principal is the amount of money deposited or borrowed.
- A tax is the amount by which the regular price of an item is reduced.
- To find a percent of increase, compare the amount of the increase to the new amount.
- If the new amount is greater than the original amount, then the percent of change is percent of decrease.

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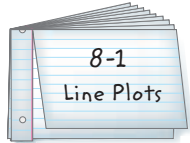


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Key Concepts

Mean, Median, and Mode (Lesson 8-2)

- The mean of a set of data is the sum of the data divided by the number of items in the set.
- The median of a set of data is the middle number of the ordered data if there is an odd number of values, or the mean of the middle two numbers if there is an even number of values.
- The mode of a set of data is the number or numbers that occur most often. If there are two or more numbers that occur most often, all of them are modes.

Statistical Displays (Lessons 8-1, 8-3 through 8-9)

- Line plots show how many times each number occurs in a data set.
- Stem-and-leaf plots list all individual numerical data in a condensed form.
- Bar graphs show the data in specific categories.
- Histograms show the frequency of data divided into smaller intervals.
- Circle graphs compare data parts to the whole.
- Line graphs show change over a period of time.
- Scatter plots determine if there is a relationship between two sets of data.

Sampling Methods (Lesson 8-8)

- In a random sample, each item or person in the population is as likely to be chosen as any other.
- A convenience sample includes members of a population that are easily accessed.
- A voluntary response sample involves only those who want to participate in the sampling.

Key Vocabulary

analyze (p. 397)	measures of central tendency (p. 402)
bar graph (p. 415)	median (p. 403)
cluster (p. 397)	mode (p. 403)
convenience sample (p. 439)	outlier (p. 397)
mean (p. 402)	random sample (p. 438)
data (p. 396)	range (p. 397)
histogram (p. 416)	scatter plot (p. 427)
leaf (p. 410)	statistics (p. 396)
line graph (p. 426)	stem (p. 410)
line plot (p. 396)	stem-and-leaf plot (p. 410)
mean (p. 402)	voluntary response sample (p. 439)

Vocabulary Check

State whether each sentence is *true* or *false*.
If *false*, replace the underlined word or number to make a true sentence.

1. The range is the difference between the greatest and the least values in a set of data.
2. The mode divides a set of data in half.
3. A graph that uses bars to make comparisons is a bar graph.
4. A scatter plot shows two sets of related data.
5. The median is a data value that is quite separated from the rest of the data.
6. The mean is the arithmetic average of a set of data.
7. The number or item that appears most often in a set of data is the mode.
8. The range is the middle number of the ordered data, or the mean of the middle two numbers.

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Key Concepts

Probability (Lesson 9-1)

- The probability of a simple event is a ratio that compares the number of favorable outcomes to the number of possible outcomes.

Fundamental Counting Principle (Lesson 9-3)

- If event M has m possible outcomes and is followed by event N that has n possible outcomes, then the event M followed by N has $m \times n$ possible outcomes.

Theoretical and Experimental

Probability (Lesson 9-7)

- Theoretical probability is based on what *should* happen when conducting a probability experiment.
- Experimental probability is based on what *actually occurred* during a probability experiment.

Independent Events (Lesson 9-8)

- The probability of two independent events can be found by multiplying the probability of the first event by the probability of the second event.

Dependent Events (Lesson 9-8)

- If two events, A and B , are dependent, then the probability of both events occurring is the product of the probability of A and the probability of B after A occurs.

Disjoint Events (Lesson 9-8)

- If two events are disjoint, then the probability that either event will occur is the sum of their individual probabilities.

Key Vocabulary

combination (p. 480)	independent events (p. 492)
complementary events (p. 462)	outcome (p. 460)
composite event (p. 492)	permutation (p. 465)
dependent events (p. 493)	probability (p. 460)
disjoint events (p. 494)	random (p. 461)
experimental probability (p. 486)	sample space (p. 465)
Fundamental Counting Principle (p. 471)	simple event (p. 460)
	theoretical probability (p. 486)
	tree diagram (p. 466)

Vocabulary Check

State whether each sentence is *true* or *false*.
If *false*, replace the underlined word or number to make a true sentence.

- Composite events consists of two or more simple events.
- A random outcome is an outcome that occurs by chance.
- $P(\text{not } A)$ is read the permutation of the complement of A .
- The Fundamental Counting Principle counts the number of possible outcomes using the operation of addition.
- Events in which the outcome of the first event does not affect the outcome of the other event(s) are simple events.
- The sample space of an event is the set of outcomes not included in the event.
- Events that cannot occur at the same time are called dependent events.

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What I Know About Polygons	What I Need to Know	What I've Learned

Key Concepts

Angles (Lessons 10-1 and 10-2)

- Two angles are adjacent if they have the same vertex, share a common side, and do not overlap.
- Two angles are vertical if they are opposite angles formed by the intersection of two lines.
- Two angles are complementary if the sum of their measures is 90° .
- Two angles are supplementary if the sum of their measures is 180° .

Triangles (Lesson 10-4)

- The sum of the measures of the angles of a triangle is 180° .

Quadrilaterals (Lesson 10-6)

- The sum of the measures of the angles of a quadrilateral is 360° .

Similar Figures (Lesson 10-7)

- If two figures are similar then the corresponding sides are proportional and the corresponding angles are congruent.

Transformations (Lessons 10-9 and 10-10)

- When translating a figure, every point in the original figure is moved the same distance in the same direction.
- When reflecting a figure, every point in the original figure is the same distance from the line of reflection as its corresponding point on the original figure.

Key Vocabulary

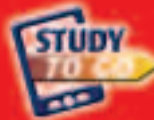
acute angle (p. 511)	pentagon (p. 546)
angle (p. 510)	polygon (p. 546)
circle graph (p. 518)	quadrilateral (p. 533)
complementary angles (p. 514)	reflection (p. 559)
congruent angles (p. 511)	regular polygon (p. 546)
congruent figures (p. 554)	rhombus (p. 533)
degrees (p. 510)	right angle (p. 511)
hexagon (p. 546)	similar figures (p. 540)
indirect measurement (p. 542)	straight angle (p. 511)
line of symmetry (p. 558)	supplementary angles (p. 514)
line symmetry (p. 558)	tessellation (p. 548)
obtuse angle (p. 511)	transformation (p. 553)
octagon (p. 546)	translation (p. 553)
parallelogram (p. 533)	trapezoid (p. 533)
	triangle (p. 524)
	vertex (p. 510)

Vocabulary Check

State whether each sentence is *true* or *false*.
If *false*, replace the underlined word or number to make a true sentence.

- Two angles whose measures add to 180° are called complementary angles.
- A hexagon is a polygon with 6 sides.
- An angle whose measure is less than 90° is called a right angle.
- The vertex is where the sides of angle meet.
- The point $(3, -2)$ when translated up 3 units and to the left 5 units becomes $(6, -7)$.
- A trapezoid has both pairs of opposite sides parallel.

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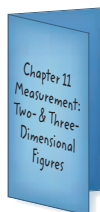


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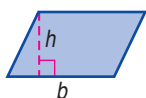
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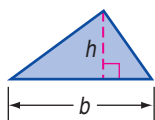
Key Concepts

Area (Lessons 11-1, 11-2, and 11-6)



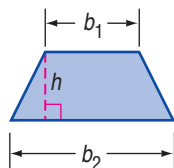
parallelogram

$$A = bh$$



triangle

$$A = \frac{1}{2}bh$$



trapezoid

$$A = \frac{1}{2}h(b_1 + b_2)$$

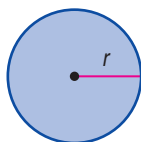
Circles (Lessons 11-3 and 11-4)

• circumference

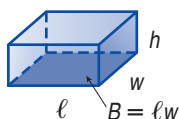
$$C = \pi d \text{ or } C = 2\pi r$$

• area

$$A = \pi r^2$$

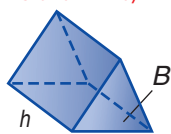


Volume (Lessons 11-9 and 11-10)

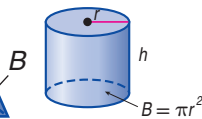


rectangular
prism

$$V = Bh$$



triangular
prism



cylinder

$$V = Bh \text{ or } \pi r^2 h$$

Key Vocabulary

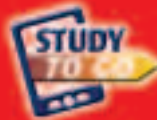
base (p. 572)	π (pi) (p. 584)
center (p. 584)	prism (p. 603)
circle (p. 584)	pyramid (p. 603)
circumference (p. 584)	radius (p. 584)
complex figure (p. 596)	rectangular prism (p. 613)
cone (p. 604)	semicircle (p. 596)
cylinder (p. 604)	sphere (p. 604)
diameter (p. 584)	three-dimensional figure (p. 603)
edge (p. 603)	triangular prism (p. 615)
face (p. 603)	vertex (p. 603)
height (p. 572)	volume (p. 613)
lateral face (p. 603)	

Vocabulary Check

Choose the correct term or number to complete each sentence.

1. A (rectangular prism, rectangle) is a three-dimensional figure that has three sets of parallel congruent sides.
2. The (volume, surface area) of a three-dimensional figure is the measure of the space occupied by it.
3. $A = \frac{1}{2}h(b_1 + b_2)$ is the formula for the area of a (triangle, trapezoid).
4. Volume is measured in (square, cubic) units.
5. A (cylinder, prism) is a three-dimensional figure that has two congruent, parallel circles as its bases.
6. The volume of a rectangular prism is found by (adding, multiplying) the length, the width, and the height.
7. The formula for the area of a (square, circle) is $A = \pi r^2$.

Study Guide and Review



Download Vocabulary
Review from ca.gr6math.com

FOLDABLES
Study Organizer

GET READY to Study

Be sure the following
Key Concepts are noted
in your Foldable.

Ch. 12	Rectangular Prisms	Cylinders
Class Concepts		
Find Words		
Find Surface Area		

Key Concepts

Irrational Numbers (Lesson 12-1)

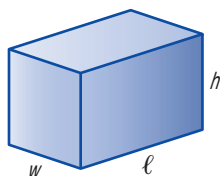
- An irrational number is a number that *cannot* be written as a fraction.

Pythagorean Theorem (Lesson 12-2)

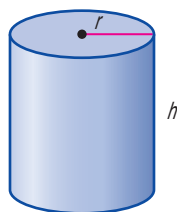
- In a right triangle, the square of the length of the hypotenuse equals the sum of the squares of the lengths of the legs.

Surface Area (Lessons 12-4, 12-5)

- The surface area S of a rectangular prism with length ℓ , width w , and height h is the sum of the areas of the faces. $S = 2\ell w + 2\ell h + 2wh$



- The surface area of S of a cylinder with height h and a radius r is the sum of the area of the circular bases and the area of the curved surface. $S = 2\pi r^2 + 2\pi rh$



Key Vocabulary

hypotenuse (p. 640) Pythagorean Theorem (p. 640)
irrational number (p. 637) surface area (p. 648)
leg (p. 640)

Vocabulary Check

State whether each sentence is *true* or *false*.
If *false*, replace the underlined word or number to make a true sentence.

- The side opposite the right angle in a scalene triangle is called a hypotenuse.
- Either of the two sides that form the right angle of a right triangle is called a hypotenuse.
- An irrational number is a number that cannot be written as a fraction.
- The Pythagorean Theorem states that in a right triangle, the square of the length of the hypotenuse equals the difference of the squares of the lengths of the legs.
- The sum of the areas of all the surfaces of a three-dimensional figure is called the surface area.
- The formula for finding the surface area of a cylinder is $S = 2\ell w + 2\ell h + 2wh$.
- Rational numbers include only positive numbers.
- The Pythagorean Theorem can be used to find length of the hypotenuse of a right triangle if the measures of both legs are known.
- To find the surface area of a rectangular prism, you must know the measurements of the height and the radius.
- The square root of a perfect square is a rational number.

Prerequisite Skills

1 Divisibility Patterns

In $54 \div 6 = 9$, the quotient, 9, is a whole number. So, we say that 54 is **divisible** by 6. You can use the following rules to determine whether a number is by 2, 3, 4, 5, 6, 9, and 10.

A number is divisible by:

- 2 if the ones digit is divisible by 2.
- 3 if the sum of the digits is divisible by 3.
- 4 if the number formed by the last two digits is divisible by 4.
- 5 if the ones digit is 0 or 5.
- 6 if the number is divisible by both 2 and 3.
- 9 if the sum of the digits is divisible by 9.
- 10 if the ones digit is 0.

EXAMPLE Use Divisibility Rules

Determine whether 972 is divisible by 2, 3, 4, 5, 6, 9, or 10.

- 2: Yes; the ones digit, 2, is divisible by 2.
 3: Yes; the sum of the digits, $9 + 7 + 2 = 18$, is divisible by 3.
 4: Yes; the number formed by the last two digits, 72, is divisible by 4.
 5: No; the ones digit is not 0 or 5.
 6: Yes; the number is divisible by 2 and 3.
 9: Yes; the sum of the digits, 18, is divisible by 9.
 10: No; the ones digit is not 0.

Exercises

Use divisibility rules to determine whether the first number is divisible by the second number.

- | | | |
|--------------|---------------|--------------|
| 1. 447; 3 | 2. 135; 6 | 3. 240; 4 |
| 4. 419; 3 | 5. 831; 3 | 6. 4,408; 4 |
| 7. 7,110; 5 | 8. 1,287; 9 | 9. 2,984; 9 |
| 10. 7,026; 6 | 11. 1,260; 10 | 12. 8,903; 6 |

Determine whether each number is divisible by 2, 3, 4, 5, 6, 9, or 10.

- | | | |
|------------|------------|-----------|
| 13. 712 | 14. 1,035 | 15. 8,901 |
| 16. 462 | 17. 270 | 18. 1,005 |
| 19. 32,221 | 20. 8,340 | 21. 920 |
| 22. 50,319 | 23. 64,042 | 24. 3,498 |

25. **MEASUREMENT** Jordan has 5,280 feet of rope. Can he cut the rope into 9-foot pieces and use all of the rope? Explain.

2 Place Value and Decimals

Our number system is based on units of ten. Numbers like 1.35, 0.8, and 25.09 are called decimals.

You can use a place-value chart like the one at the right to help you read decimals and write them in words. Use the word *and* to represent the decimal point.

10,000	1,000	100	10	1						
ten thousands	thousands	hundreds	tens	ones	decimal point	tenths	hundredths	thousandths	ten-thousandths	hundred-thousandths
			2	5	.	0	9			

EXAMPLES Write Decimal in Words

Write each decimal in words

1 20.09

tens	ones	decimal point	tenths	hundredths
2	0	.	0	9

twenty and nine hundredths

2 6.738

ones	decimal point	tenths	hundredths	thousandths
6	.	7	3	8

six and seven hundred thirty-eight thousandths

You can also write decimals in **expanded notation** using place value and decimals or their fraction equivalents as shown at the right.

Decimal	0.1	0.01	0.001	0.0001
Fraction	$\frac{1}{10}$	$\frac{1}{10^2}$	$\frac{1}{10^3}$	$\frac{1}{10^4}$

EXAMPLE Write a Decimal in Expanded Notation

3 Write 2.814 in expanded notation using decimals and using fractions.

Write the product of each digit and its place value.

$$2.814 = (2 \times 1) + (8 \times 0.1) + (1 \times 0.01) + (4 \times 0.001)$$

$$2.814 = (2 \times 1) + \left(8 \times \frac{1}{10}\right) + \left(1 \times \frac{1}{10^2}\right) + \left(4 \times \frac{1}{10^3}\right)$$

Replace the decimals with their fraction equivalents.

Exercises

Write each decimal in words.

1. 6.37

2. 13.7

3. 145.66

4. 92.03

5. 4,280.899

6. 0.6

7. 0.69

8. 0.00012

9. 7.005

10. 2000.0002

Write each decimal in expanded notation using decimals and using fractions.

11. 0.8

12. 5.3

13. 6.79

14. 9.132

15. 65.002

16. 0.625

17. 0.0072

18. 100.001

19. 548.845

20. 4,260.705

21. **BATS** The northern blossom bat is one of the world's smallest bats. It weighs just 0.53 ounce. Write its weight in expanded notation using fractions.

3 Comparing and Ordering Decimals

To determine which of two decimals is greater, you can compare the digits in each place-value position, or you can use a number line.

EXAMPLES Order Decimals

1 Which is greater, 7.4 or 7.63?

METHOD 1 Use place value.

7.4 Line up the decimal points.

7.63 Starting at the left, compare the digits in each place-value position.

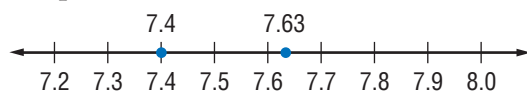


The digits in the tenths place are not the same.

Since 6 tenths $>$ 4 tenths, $7.63 > 7.4$.

METHOD 2 Use a number line.

Graph each decimal on a number line and compare.



On a number line, numbers to the right are greater than numbers to the left.

7.63 is to the right of 7.4, so $7.63 > 7.4$.

2 Order 12.15, 12.08, and 12.103 from least to greatest.

12.15 Line up the decimal points.

12.08 12.08 is less than 12.15 and 12.103 since $0 < 1$ in the tenths place.

12.103 12.103 is less than 12.15 since $0 < 5$ in the hundredths place.

So, the numbers from least to greatest are 12.08, 12.103, and 12.15.

Exercises

Replace each $\bullet$ with $<$, $>$, or $=$ to make a true sentence.

- | | | |
|-------------------------------|----------------------------|-------------------------|
| 1. $1.22 \bullet 1.02$ | 2. $0.97 \bullet 1.06$ | 3. $7.90 \bullet 7.9$ |
| 4. $1.3 \bullet 1.31$ | 5. $4.03 \bullet 4.01$ | 6. $0.77 \bullet 0.69$ |
| 7. $0.8 \bullet 0.08$ | 8. $0.68 \bullet 0.680$ | 9. $3.28 \bullet 3.279$ |
| 10. $25.23 \bullet 25.32$ | 11. $77.55 \bullet 77.65$ | 12. $1.29 \bullet 1.43$ |
| 13. $310.36 \bullet 310.3600$ | 14. $0.0034 \bullet 0.034$ | 15. $9.09 \bullet 9$ |

Order each group of decimals from least to greatest.

- | | |
|--------------------------------|------------------------------------|
| 16. 5.13, 5.07, 5.009 | 17. 7.9, 7.088, 8.02, 7.98 |
| 18. 0.087, 0.901, 2, 1.001 | 19. 12.3, 12.008, 12.54, 12 |
| 20. 60.5, 60.05, 60.55, 60.505 | 21. 505.9, 505.09, 505.91, 505.99 |
| 22. 0.02, 0.2, 0.022, 0.002 | 23. 0.99, 0.90, 0.09, 0.999, 0.099 |

4 Rounding Decimals

When a number has more decimal places than you want or need, you can **round** it. Use the rules below to round a number to any place value.

Rounding Decimals

- Look at the digit to the right of the place being rounded.
- If the digit is 4 or less, the digit being rounded remains the same.
 - If the digit is 5 or greater, the digit being rounded is rounded up.

EXAMPLES Round Decimals

1 Round 3.92 to the nearest tenth.

tenths place



3.92 Look at the digit to the right of the tenths place. Since 2 is less than 5, the digit in the tenths place stays the same.

3.92 rounded to the nearest tenth is 3.9.

2 Round 46.297 to the nearest hundredth.

hundredths place



46.297 Look at the digit to the right of the hundredths place. Since 7 is greater than 5, round 9 up.

46.297 rounded to the nearest hundredth is 46.30.

Exercises

Round each number to the given place value.

- | | | |
|-------------------------|----------------------|---------------------|
| 1. 0.315; tenth | 2. 0.2456; hundredth | 3. 17.499; tenth |
| 4. 43.219; hundredth | 5. 15.522; tenth | 6. 9.6; ones |
| 7. 7.0375; thousandth | 8. 16.399; tenth | 9. 6.95; tenth |
| 10. 99.1283; thousandth | 11. 1,000.37; tenth | 12. 750.523; ones |
| 13. 0.445; hundredth | 14. 490.299; ones | 15. 999.99; tenth |
| 16. 476.835; hundredth | 17. 682.596; tenth | 18. 1,000.562; ones |

Round each number to the underlined place-value position.

- | | | |
|---------------------|----------------------|----------------------|
| 19. 0. <u>7</u> 89 | 20. 0. <u>9</u> 6 | 21. 1.5 <u>7</u> 246 |
| 22. 2 <u>3</u> .48 | 23. 1.7 <u>0</u> 4 | 24. 0. <u>1</u> 63 |
| 25. 15. <u>4</u> 51 | 26. 4.52 <u>9</u> 88 | 27. 0. <u>7</u> 87 |
| 28. 3 <u>8</u> .56 | 29. 5 <u>9</u> .61 | 30. <u>0</u> .555 |
| 31. 0. <u>5</u> 55 | 32. 0.5 <u>5</u> 5 | 33. 10. <u>9</u> 29 |
| 34. 10.9 <u>2</u> 9 | 35. 42 <u>5</u> .599 | 36. 0.929 <u>2</u> 9 |

37. **HORSES** A trainer recorded a racehorse running at 47.54 miles per hour. Round this speed to the nearest mile per hour.
38. **POPULATION** In 2000, the population of Texas was 20.85 million. Round to the nearest million.

5 Estimating with Decimals

Estimation can be used to provide quick answers when an exact answer is not necessary. It is also an excellent way to check whether your answer is reasonable. One method of estimating is to use rounding. Round numbers to any place value that makes estimation easier.

EXAMPLES Estimate by Rounding

Estimate by rounding.

1 $23.485 - 9.757$

$$\begin{array}{r} 23.485 \rightarrow 23 \\ - 9.757 \rightarrow -10 \\ \hline 13 \end{array}$$

Round to the nearest whole numbers.

The difference is about 13.

2 $6.43 + 2.17 + 9.1 + 4.87$

$$\begin{array}{r} 6.43 \rightarrow 6 \\ 2.17 \rightarrow 2 \\ 9.1 \rightarrow 9 \\ + 4.87 \rightarrow +5 \\ \hline 22 \end{array}$$

Round to the nearest whole numbers.

The sum is about 22.

Another way to estimate sums is to use **clustering**. This strategy is used when all the numbers are close to a common value.

EXAMPLE Estimate by Clustering

3 Estimate $9.775 + 9.862 + 9.475 + 9.724$ by clustering.

All of the numbers are clustered around 10. There are four numbers.

So, the sum is about 4×10 or 40.

Exercises

Estimate by rounding.

- | | | |
|------------------|---------------------|---------------------|
| 1. $8.56 + 5.34$ | 2. $34.84 - 17.69$ | 3. $6.8 + 2.4$ |
| 4. $40.79 - 6.8$ | 5. $6.9 + 5.2$ | 6. $23.84 + 12.13$ |
| 7. $34.3 - 18.9$ | 8. $7.5 + 8.4$ | 9. $65.48 - 9.3$ |
| 10. $26.3 + 9.7$ | 11. $33.21 - 8.23$ | 12. $67.86 - 24.35$ |
| 13. $8.99 - 2.6$ | 14. $121.5 + 487.8$ | 15. $32.5 + 81.4$ |

Estimate by clustering.

- | | |
|--|---|
| 16. $18.4 + 22.5 + 20.7$ | 17. $56.9 + 63.2 + 59.3 + 61.1$ |
| 18. $42.3 + 41.5 + 39.8 + 40.4$ | 19. $77.8 + 75.6 + 81.2 + 79.9$ |
| 20. $239.8 + 242.43 + 236.20 + 240.77$ | 21. $9.9 + 10.0 + 10.3 + 11.1 + 9.8 + 11.2$ |
| 22. $50.4 + 51.1 + 48.9 + 49.5 + 50.8$ | 23. $100.5 + 97.8 + 101.6 + 100.2 + 99.3$ |

6 Adding and Subtracting Decimals

To add or subtract decimals, write the numbers in a column and line up the decimal points. In some cases, you may want to *annex*, or place zeros at the end of the decimals, to help align the columns. Then add or subtract as with whole numbers and bring down the decimal point. Always estimate first to see whether your answer is reasonable.

EXAMPLES Add and Subtract Decimals

Find each sum or difference.

1 $12.6 + 4.5$ **Estimate** $13 + 5 = 18$

$$\begin{array}{r} 12.6 \\ + 4.5 \\ \hline 17.1 \end{array}$$

Line up the decimal points.
Add.
The sum is close to the estimate.

2 $42.17 - 15.85$ **Estimate** $42 - 16 = 26$

$$\begin{array}{r} 42.17 \\ - 15.85 \\ \hline 26.32 \end{array}$$

Line up the decimal points.
Subtract.
The difference is close to the estimate.

3 $57.125 + 7.63$ **Estimate** $57 + 8 = 65$

$$\begin{array}{r} 57.125 \\ + 7.630 \\ \hline 64.755 \end{array}$$

Annex a zero to align the columns.
Compare to the estimate.

4 $25 - 15.25$ **Estimate** $25 - 15 = 10$

$$\begin{array}{r} 25.00 \\ - 15.25 \\ \hline 9.75 \end{array}$$

Annex a zero to align the columns.
Compare to the estimate.

Exercises

Find each sum or difference.

1. 0.132
 $- 0.021$

2. 3.78
 $+ 0.21$

3. 13.2
 $+ 12.8$

4. 5.86
 $- 1.51$

5. 42.07
 $- 38.78$

6. 14.7
 $+ 351.82$

7. 42.3
 $+ 0.81$

8. 12.3
 $- 0.847$

9. 342.9
 $- 0.18$

10. 282.45
 $- 111.3$

11. 100
 $- 0.48$

12. 82.23
 $+ 0.88$

13. 128.01
 $- 39.117$

14. 80.05
 $- 79.06$

15. 104.98
 $- 0.12$

16. $0.42 + 0.68$

17. $0.48 + 2.901$

18. $5.8 + 3.92$

19. $38.63 + 38.63$

20. $8 - 2.54$

21. $16.354 - 0.2$

22. $0.125 + 0.78$

23. $8.2 - 6.9$

24. $1.245 + 3.842$

25. $3.2 + 1.23$

26. $0.889 - 0.3$

27. $22.22 + 1.475$

28. $10 - 0.25$

29. $33.16 - 0.08$

30. $1.254 + 0.5$

31. $44.698 - 14.903$

32. $10 - 0.005$

33. $722.86 + 0.024$

34. $100.211 + 8.004$

35. $86.124 + 32.822$

36. $6.9 + 1.1$

37. $75 - 0.24$

38. $13 - 0.324$

39. $0.8 + 1.2$

40. What is the sum of 35.009 and 3.6?

41. Find 9.1 minus 5.625.

42. **AIR TRAVEL** The numbers of passengers, in millions, passing through four airports in a recent year were 80.2, 72.1, 68.5, and 60.7. Find the total number of passengers for the four airports.

7 Multiplying Decimals

To multiply decimals, multiply as with whole numbers. The product has the same number of decimal places as the sum of the decimal places of the factors. Use estimation to determine whether your answers are reasonable.

EXAMPLES Multiply Decimals

Multiply.

1 1.3×0.9 **Estimate** $1 \times 1 = 1$

$$\begin{array}{r} 1.3 \quad \leftarrow 1 \text{ decimal place} \\ \times 0.9 \quad \leftarrow 1 \text{ decimal place} \\ \hline 1.17 \quad \leftarrow 2 \text{ decimal places} \end{array}$$

The product is reasonable.

2 0.054×1.6 **Estimate** $0 \times 2 = 0$

$$\begin{array}{r} 0.054 \quad \leftarrow 3 \text{ decimal places} \\ \times 1.6 \quad \leftarrow 1 \text{ decimal place} \\ \hline 324 \\ 540 \end{array}$$

0.0864 Annex a zero on the left so the answer has four decimal places. **Compare to the estimate.**

Exercises

Place the decimal point in each product. Add zeros if necessary.

1. $1.32 \times 4 = 528$

2. $0.07 \times 1.1 = 77$

3. $0.4 \times 0.7 = 28$

4. $1.9 \times 0.6 = 114$

5. $1.4 \times 0.09 = 126$

6. $5.48 \times 3.6 = 19728$

7. $4.5 \times 0.34 = 153$

8. $0.45 \times 0.02 = 9$

9. $150.2 \times 32.75 = 4919050$

Multiply.

10. $\begin{array}{r} 0.2 \\ \times 6 \\ \hline \end{array}$

11. $\begin{array}{r} 0.3 \\ \times 0.9 \\ \hline \end{array}$

12. $\begin{array}{r} 0.45 \\ \times 0.12 \\ \hline \end{array}$

13. $\begin{array}{r} 0.0023 \\ \times 32 \\ \hline \end{array}$

14. $\begin{array}{r} 1.5 \\ \times 2.7 \\ \hline \end{array}$

15. $\begin{array}{r} 10.1 \\ \times 9 \\ \hline \end{array}$

16. $\begin{array}{r} 2 \\ \times 0.3 \\ \hline \end{array}$

17. $\begin{array}{r} 6.78 \\ \times 1.3 \\ \hline \end{array}$

18. $\begin{array}{r} 200 \\ \times 0.004 \\ \hline \end{array}$

19. $\begin{array}{r} 0.0023 \\ \times 0.35 \\ \hline \end{array}$

20. 15.8×11

21. 88×2.5

22. 33×0.03

23. 36×0.46

24. 0.003×482

25. 1.88×1.11

26. 0.6×2

27. 38.3×29.1

28. 0.7×18

29. 8×0.3

30. 12.2×12.4

31. 380×1.25

32. 42×0.17

33. 0.4×16

34. 0.23×0.2

35. 0.44×0.5

36. 0.44×55

37. 44×0.55

38. **JOBS** Antonia earns \$10.75 per hour. What are her total weekly earnings if she works 34.5 hours? Round to the nearest cent.

8 Powers of Ten

You can use a pattern to mentally find the product of any number and a power of 10 that is greater than 1. Count the number of zeros in the power of 10 or use the exponent. Then move the decimal point that number of places to the right.

Decimal Power of 10	Product
19.7×10^1 (or 10)	= 197
19.7×10^2 (or 100)	= 1,970
19.7×10^3 (or 1,000)	= 19,700
19.7×10^4 (or 10,000)	= 197,000

EXAMPLES Use Mental Math to Multiply

Multiply mentally.

1 12.562×100

$$12.562 \times 100 = 12.562$$

$$= 1,256.2$$

Move the decimal point two places to the right, since 100 has two zeros.

2 0.59×10^4

$$0.59 \times 10^4 = 0.5900$$

$$= 5,900$$

Move the decimal point four places to the right, since the exponent is 4.

To mentally multiply by a power of ten that is less than 1, count the number of decimal places. Or, if the power is written as a fraction, use the exponent in the denominator. Then move the decimal point that number of places *to the left*.

Decimal Power of 10	Product
19.7×0.1 (or $\frac{1}{10^1}$)	= 1.97
19.7×0.01 (or $\frac{1}{10^2}$)	= 0.197
19.7×0.001 (or $\frac{1}{10^3}$)	= 0.0197

EXAMPLES Use Mental Math to Multiply

Multiply mentally.

3 10.5×0.01

$$10.5 \times 0.01 = 10.5$$

$$= 0.105$$

Move the decimal point two places to the left.

4 $5,284 \times 0.00001$

$$5,284 \times 0.00001 = 05284$$

$$= 0.05284$$

Move the decimal point five places to the left.

Exercises

Multiply mentally.

- 12.53×10
- 4.6×10^3
- 78.4×0.01
- 0.05×100
- 4.527×100
- $2.78 \times 1,000$
- 13.58×0.01
- 5.49×10^3
- 0.1×0.8
- 0.925×10
- 99.44×10^2
- 0.01×16
- 1.32×10^3
- $0.56 \times 10,000$
- 1.4×0.001
- 11.23×10^5
- 68.94×0.01
- 0.8×10^4
- 28.1×0.01
- 9.3×10^7
- $625,799 \times 0.0001$

9 Dividing Decimals

To divide two decimals, use the following steps.

- If necessary, change the divisor to a whole number by moving the decimal point to the right. **You are multiplying the divisor by a power of ten.**
- Move the decimal point in the dividend the same number of places to the right. **You are multiplying the dividend by the same power of ten.**
- Divide as with whole numbers.

EXAMPLES Divide Decimals

Divide.

$$25.8 \div 2 \quad \text{Estimate } 26 \div 2 = 13$$

1

$$\begin{array}{r} 12.9 \\ 2 \overline{)25.8} \\ \underline{-2} \\ 5 \\ \underline{-4} \\ 18 \\ \underline{-18} \\ 0 \end{array}$$

The divisor, 2, is already a whole number, so you do not need to move the decimal point. Divide as with whole numbers. Then place the decimal directly above the decimal point in the dividend.

Compared to the estimate, the quotient, 12.9, is reasonable.

2

$$199.68 \div 9.6 \quad \text{Estimate } 200 \div 10 = 20$$

$$\begin{array}{r} 20.8 \\ 9.6 \overline{)199.68} \\ \underline{-192} \\ 768 \\ \underline{-768} \\ 0 \end{array}$$

Move each decimal point one place to the right.

Compare the answer to the estimate.

Exercises

Divide.

- | | | | | |
|------------------------------|------------------------------|-----------------------------|------------------------------|------------------------------|
| 1. $0.3 \overline{)9.81}$ | 2. $12 \overline{)0.12}$ | 3. $3.2 \overline{)5.76}$ | 4. $0.22 \overline{)0.0132}$ | 5. $0.04 \overline{)0.008}$ |
| 6. $3.18 \overline{)0.636}$ | 7. $0.2 \overline{)8.24}$ | 8. $82.3 \overline{)823}$ | 9. $12.02 \overline{)24.04}$ | 10. $0.5 \overline{)85}$ |
| 11. $74.9 \overline{)5.992}$ | 12. $19.2 \overline{)4.416}$ | 13. $1.9 \overline{)38.57}$ | 14. $13.8 \overline{)131.1}$ | 15. $6.48 \overline{)259.2}$ |
| 16. $812 \div 0.4$ | 17. $0.34 \div 0.2$ | 18. $14.4 \div 0.12$ | 19. $90.175 \div 2.5$ | |
| 20. $39.95 \div 799$ | 21. $88.8 \div 444$ | 22. $613.8 \div 66$ | 23. $2,445.3 \div 33$ | |
| 24. $20.24 \div 2.3$ | 25. $45 \div 0.09$ | 26. $2.475 \div 0.03$ | 27. $4.6848 \div 0.366$ | |
| 28. $180 \div 0.36$ | 29. $97.812 \div 1.1$ | 30. $23 \div 0.023$ | 31. $1,680.042 \div 44.2$ | |

32. **OLYMPICS** In the 2000 Olympics, Michael Johnson of the U.S. ran the 400-meter run in 43.84 seconds. To the nearest hundredth, find his speed in meters per second.
33. **EARTH SCIENCE** It takes Pluto 247.69 Earth years to revolve once around the Sun. It takes Jupiter 11.86 Earth years to revolve once around the Sun. About how many times longer does it take Pluto than Jupiter to revolve once around the Sun?

10 Mixed Numbers and Improper Fractions

A **mixed number** is the sum of a whole number and a fraction. An **improper fraction** is a fraction with a numerator that is greater than or equal to the denominator.

mixed number: $8\frac{1}{2}$ improper fraction: $\frac{17}{2}$

To write a mixed number as an improper fraction, use the following steps.

- Multiply the whole number by the denominator.
- Add the numerator.
- Write the sum as the numerator of the improper fraction.

EXAMPLES Write Mixed Numbers as Improper Fractions

Write each mixed number as an improper fraction.

1 $6\frac{1}{2}$ $6\frac{1}{2} = \frac{(6 \cdot 2) + 1}{2}$ $\leftarrow$ Multiply 6 by 2. Add 1. $\leftarrow$ The denominator is 2. $= \frac{12 + 1}{2}$ or $\frac{13}{2}$ Simplify.	2 $3\frac{4}{5}$ $3\frac{4}{5} = \frac{(3 \cdot 5) + 4}{5}$ $\leftarrow$ Multiply 3 by 5. Add 4. $\leftarrow$ The denominator is 5. $= \frac{15 + 4}{5}$ or $\frac{19}{5}$ Simplify.
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To write an improper fraction as a mixed number, divide the numerator by the denominator. Write the remainder as the numerator of the fraction.

EXAMPLE Write Improper Fractions as Mixed Numbers

3 Write $\frac{7}{4}$ as a mixed number.

$7 \div 4 = 1 \text{ R } 3$ Divide the numerator by the denominator.

$= 1\frac{3}{4}$ Write the remainder as the numerator of the fraction.

So, $\frac{7}{4} = 1\frac{3}{4}$.

Exercises

Write each mixed number as an improper fraction.

- | | | | | |
|--------------------|---------------------|--------------------|---------------------|---------------------|
| 1. $2\frac{1}{3}$ | 2. $1\frac{5}{8}$ | 3. $4\frac{5}{7}$ | 4. $3\frac{3}{4}$ | 5. $9\frac{1}{3}$ |
| 6. $1\frac{1}{10}$ | 7. $10\frac{2}{3}$ | 8. $12\frac{1}{4}$ | 9. $6\frac{3}{8}$ | 10. $11\frac{1}{2}$ |
| 11. $2\frac{3}{8}$ | 12. $20\frac{3}{5}$ | 13. $5\frac{7}{8}$ | 14. $27\frac{1}{3}$ | 15. $5\frac{2}{3}$ |

Write each improper fraction as a mixed number.

- | | | | | |
|--------------------|--------------------|---------------------|---------------------|--------------------|
| 16. $\frac{15}{2}$ | 17. $\frac{9}{7}$ | 18. $\frac{15}{8}$ | 19. $\frac{100}{3}$ | 20. $\frac{28}{5}$ |
| 21. $\frac{3}{3}$ | 22. $\frac{19}{8}$ | 23. $\frac{17}{10}$ | 24. $\frac{51}{3}$ | 25. $\frac{17}{6}$ |
| 26. $\frac{5}{3}$ | 27. $\frac{99}{3}$ | 28. $\frac{25}{7}$ | 29. $\frac{46}{9}$ | 30. $\frac{27}{4}$ |

11 Measuring and Drawing Angles

An angle is formed by two rays with a common endpoint, called a **vertex**. The measure of the “tilt” of one side of the angle compared to the other can be given in degrees measured with a **protractor**. Angles can be classified by their measure.

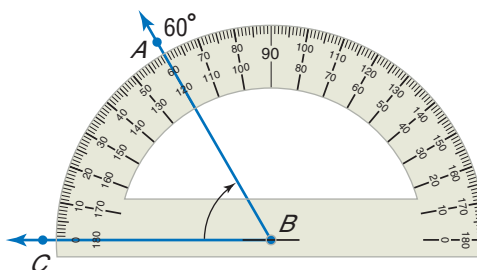
- Angles with measures less than 90° are called **acute** angles.
- Angles with measures of exactly 90° are called **right** angles.
- Angles with measures between 90° and 180° are called **obtuse** angles.
- Angles with measures of exactly 180° are called **straight** angles.

EXAMPLE Measure an Angle

- 1** Use a protractor to measure $\angle ABC$. State whether the angle is acute, right, obtuse, or straight.

Step 1 Place the center point of the protractor's base on vertex B . Align the straight side with side $\overrightarrow{BC}$ so that the marker for 0° is on side $\overrightarrow{BC}$.

Step 2 Use the scale that begins with 0° at $\overrightarrow{BC}$. Read where the other side of the angle, $\overrightarrow{BA}$, crosses this scale.



The measure of angle ABC is 60° . Using symbols, $m\angle ABC = 60^\circ$. Since $m\angle ABC < 90^\circ$, the angle is acute.

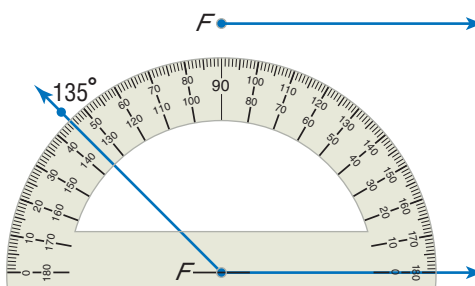
EXAMPLE Draw an Angle

- 2** Draw $\angle F$ having a measure of 135° .

Step 1 Draw a ray. Label the endpoint F .

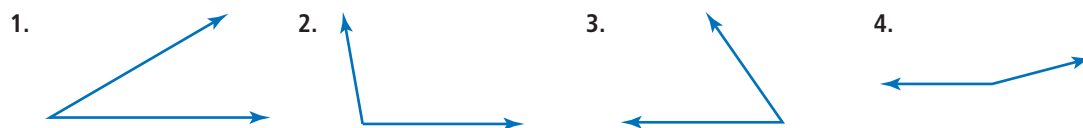
Step 2 Place the center point of the protractor's base on point F . Align the mark labeled 0 with the ray.

Step 3 Use the scale that begins with 0. Locate the mark labeled 135. Then draw the other side of the angle.



Exercises

Measure the following angles and state whether they are acute or obtuse.



Draw the following angles using a protractor.

5. 45° 6. 120° 7. 76° 8. 153°